Bounded Queues Before Learned Placement: Autonomous Management of a UAV-Borne 5G Cell for Search-and-Rescue Robots

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***Abstract*—Autonomous management of a 5G cell carried by an unmanned aerial vehicle (UAV) must keep search-and-rescue robots commandable without an operator and without knowing the terrain. Over an OpenAirInterface and Open5GS testbed serving four robots, we built and flew a three-loop management stack: a per-robot uplink cap (every 0.2 s), a liveness monitor (every 20 s), and a terrain-blind learning layer that forecasts link failure and places the UAV in three dimensions (every 30 s). The decisive loop was the simplest. Long forest outages were not coverage loss but uplink queuing inside the user equipment (UE): constant-rate video exceeded what the links drained and held command-and-control (C2) traffic minutes behind it. The cap, driven only by round-trip time and bytes sent, raised forest C2 availability from 66.8 % to 91.8 % in a patrol test and cut the median C2 outage from 143 s to 17 s in a placement run. A second silent failure, dead UE processes, had contaminated 12.4 % of the training data. With both handled, the learning layer forecasts C2 failure 20 s ahead with modest skill that falls as model capacity grows, and neither a linear nor a gradient-boosted model recovers how terrain changes the effect of altitude. Over repeated missions with the cap, no learned placement beat fixed hover: in open field every arm was within 5 points of it, and in forest the arms that moved the UAV trailed it. For autonomous management of such cells, sound inner loops, with bounded queues and verified liveness, must come before learned placement.**

I. Introduction

After an earthquake, flood or fire, the ground robots sent to search a disaster site need a mission-critical link for command and control (C2) and video, and the terrestrial network they would use is often gone or overloaded [1]. A 5G base station (gNB) carried by an unmanned aerial vehicle (UAV) can bring a private cell to the robots [2], [3]. Such a cell has to be managed: something must decide where to fly it, notice when a robot is about to lose its link, and keep the links usable under the load the robots put on them. On site there is no operator to tune it and no reliable map of terrain that may have just changed, so the management has to be autonomous, proactive and blind to the terrain, which is the setting of zero-touch management [4].

Our earlier testbed study [5] showed that terrain decides whether moving such a node helps: an uninformed roaming node raised a weak robot's C2 availability in open field and cost about 22 points in forest. The natural next step is a learning-based management layer that forecasts link failures and places the UAV in three dimensions without being told the terrain. This paper builds that layer as part of a three-loop management stack (Fig. 1), flies it in closed loop over real 5G software, and asks what each loop delivers. The answer turned out to depend less on the learning than on whether the managed system was measured and loaded soundly.

We test four claims, each against an explicit control (Section IV):

- **C1** Long forest outages under robot video are caused by uplink queuing inside the user equipment (UE), not by coverage, and a delay-adaptive uplink cap at each robot removes them without starving strong links.
- **C2** A terrain-blind model forecasts the onset of C2 failure 20 s ahead on an unseen terrain better than chance, and higher-capacity models do worse.
- **C3** Terrain-blind models do not recover how terrain changes the effect of altitude, so they cannot choose altitude.
- **C4** Learned placement does not beat fixed hover, over repeated missions in two terrains.

C1 holds; C2 and C4 hold weakly; C3 holds as a negative result.

Contributions. The work falls under wireless and 5G networks, mission-critical networks, proactive and autonomic management, artificial intelligence (AI)-based management, monitoring and measurement, and testbed experimentation. Beyond the testbed and terrain comparison of [5] (Table I), it contributes:

1) *A diagnosis and an autonomic fix.* The forest outages are a silent queueing failure of the managed system. A per-robot cap driven only by round-trip time and bytes sent, with no change to the radio access network (RAN), shortens outages from minutes to seconds and lifts both placement arms flown with and without it (Section V-A).
2) *A liveness monitor with a measurement-integrity check.* A second silent failure, the UE process dying after radio-link failure, contaminated 12.4 % of training observa-

tions; we give a mechanistic detection rule and validate it against the monitor's own record (Section III-B).

3) *An evaluation of terrain-blind learning for proactive management* on held-out terrain and in closed loop, with the negative results stated (Sections V-B and V-C).
4) *A closed-loop account of learned placement*, showing why a single mission is too short to learn the forest's altitude effect and why a cautious learner reduces to hover (Section V-D).

All results come from emulation over real 5G software; a hardware trial is the next step.

II. Background and Related Work

Aerial base stations and their testbeds. The elevation-dependent line-of-sight model [6] underlies most work on where to place a UAV base station, from 3-D placement [7], [8] and trajectory design [9] to deep reinforcement learning [10]; surveys cover the wider field [2], [3]. These works optimise coverage or rate under an analytic or simulated channel. Testbeds now run OpenAirInterface on UAVs with O-RAN control [11] or emulate 5G mobility with ray tracing [12]. Our earlier study [5] coupled a real 5G stack to a terrain simulator and found that terrain decides whether moving the node helps.

Learning-based and autonomous management. Zero-touch management aims at fully autonomous operation under human oversight [4], and O-RAN exposes closed control loops at sub-second to minute time scales in which learned controllers can run [13]; our three loops sit at the same three scales, but outside the RAN. Machine learning for link-quality estimation [14], fault management [15] and failure prediction in 5G RANs, with deep and transformer models [16], [17], learns from the history of a deployed network. A UAV arriving at a disaster site has no such history for its terrain, so we test on an unseen terrain and in closed loop.

Buffering in cellular links. Buffers that hold rather than drop packets inflate delay [18], and per-user cellular buffers are a known source [19]. Remedies range from active queue management [20] and delay-aware rate control [21] to, recently, RAN-side congestion marking: in 5G, time spent in the radio link control (RLC) buffer dominates one-way delay [22], retransmission and uplink-scheduling buffers can add delay comparable to congestion [23], and sizing the RLC buffer dynamically cuts that delay [24]. Those remedies change network queues, the RAN or the transport protocol. Our cap applies the classic additive-increase, multiplicative-decrease (AIMD) rule [25] to a delay signal at the robot, because constant-rate robot video does not back off and the UE's buffer is not ours to change.

III. System and Method

The management stack has three loops at three time scales (Fig. 1): a per-robot uplink cap that reacts within a second (Section III-D), a liveness monitor that checks every 20 s that each UE process is alive (Section III-B), and a terrain-blind learning layer that decides every 30 s where the UAV goes (Section III-C). The inner loops keep the managed system sound; the outer loop optimises it.

TABLE I
This Paper Versus Our Earlier Study [5]

	SATNAC 2026	This paper
Robots	2	4
UAV control	fixed hover vs. random walk	forecaster, planner, altitude learners vs. hover and centroid
Management loops	rule-based coverage guard	uplink cap, liveness monitor, learned x–y–z placement
Traffic	2-s 8 Mb/s bursts per cycle	continuous 1.5 Mb/s video per robot
C2 available	$\geq$1 of 5 echoes answered	ping loss $< 50\,\%$
Validation	static vs. roaming phase, one run per terrain	held-out terrain, null controls, repeated seeds
Main finding	mobility helps in open field, hurts in forest	bound queues and check liveness first; learning ties simple rules

A. Platform

The managed system is the protocol-in-the-loop testbed of [5], extended from two to four robots (Fig. 1). An OpenAirInterface (OAI) [26] gNB and four OAI UEs run over the OAI radio simulator and register to an Open5GS [27] core (OAI commit 09c9bb998f). The cell uses band n78 at 3.32 GHz with 106 resource blocks at 30 kHz spacing (40 MHz) and a 5-ms time-division pattern of five downlink and four uplink slots; every simulated channel is additive white Gaussian noise (AWGN), so there is no multipath. Each UE's virtual network (TUN) interface sits in its own Linux network namespace, so robot traffic crosses the real 5G user plane. Gazebo moves the UAV and four ground robots; a bridge reads their poses every 0.5 s and writes a terrain-specific air-to-ground path loss into each UE's channel: free-space loss with an elevation-dependent line-of-sight term [6], plus foliage [28], diffraction [29] or building loss by terrain, and correlated shadowing [30] (details in [5]). A watchdog relaunches a UE whose process exits after radio-link failure.

B. Liveness Monitor and Measurement Integrity

In OAI, a radio-link failure terminates the UE process, after which the robot's C2 stays down whatever the UAV does. The liveness monitor probes each robot's C2 every 20 s. After 90 s of total C2 loss it checks the uplink from inside the robot's namespace and, if the UE is gone, relaunches that UE alone, giving up after three failed attempts.

The same failure contaminates data recorded before the monitor worked. Once a UE process dies, the gNB's per-UE counters freeze, and the cycles are logged as link failures although they record a dead process; a model with lagged features spots them trivially (the change in block error rate is exactly zero). We flag a cycle as dead when, for at least three consecutive cycles, the gNB counters `dl_bler`, `ul_nb_rbs` and `ul_bits_per_sym` are bit-identical to the previous cycle *and* C2 is down. Frozen counters on a working link are a quiet cycle, and a failed cycle with moving counters is

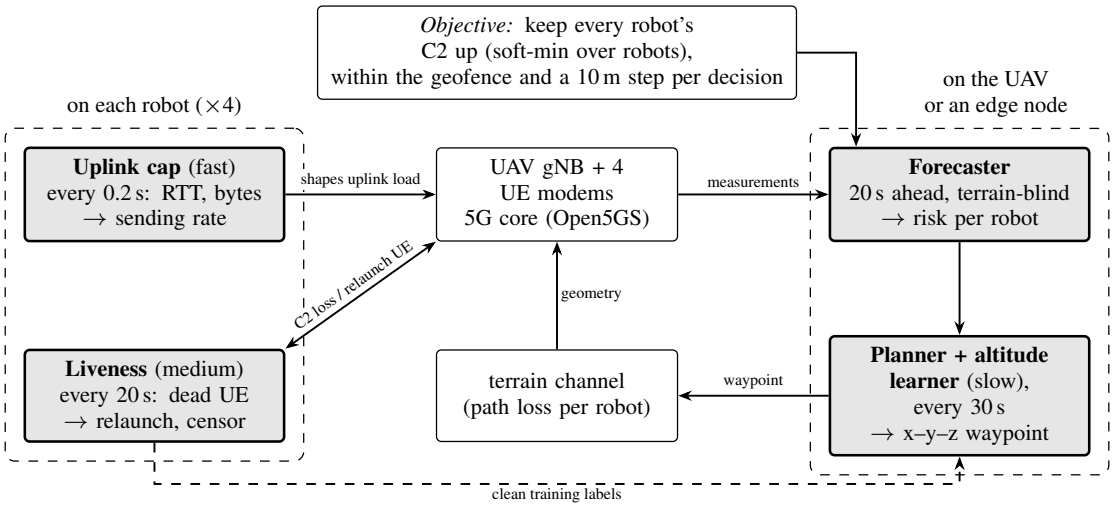


Fig. 1. Management architecture, deployment view: three loops at three time scales around the managed cell. The cap and the liveness monitor would run on each robot, the learning layer on the UAV or an edge node; in the testbed all run on one host. The liveness monitor also cleans the labels the learning layer is trained on.

a real degradation, so both conditions are required. The rule flags 12.4 % of all cycles and 72.2 % of hills failure labels (Table V), and flagged cycles are censored. Checked against the monitor's own record of dead or relaunching UEs, in the 29 of 80 robot series where the monitor was running, 85 % of the rule's flags agree, but the rule finds only 56 % of the dead cycles the monitor saw, so some contamination remains in the older data.

C. Terrain-Blind Learning Layer

Forecaster. The label is the onset of C2 failure: for a robot whose C2 is up at time t, is it down at $t+20$ s? C2 is up when fewer than 50 % of its probes are lost. Features are UAV–robot geometry, gNB-side link adaptation and error statistics, and the last commanded move, each with lags, differences and rolling statistics within one robot's series (65 features). Three inputs are forbidden: the simulator's injected path loss, the computed received power (algebraically equivalent to it), and any terrain label. A guard function checks every feature matrix before it reaches an estimator. Models are trained leave-one-terrain-out, with thresholds and calibration chosen on an inner time split of the training terrains.

Planner. Each 30-s decision, the planner scores candidate UAV positions (absolute altitudes 25–55 m in 5 m steps, crossed with lateral offsets of 0 or 15 m) by a soft minimum over the robots' predicted risks, $J(u) = \beta^{-1} \log \sum_i e^{\beta r_i(u)}$ with $\beta = 10$, so the worst-served robot dominates. It moves toward the minimiser at most 10 m in altitude per decision. The planner's risk model is the same logistic regression with spline terms on altitude, distances and elevation angle and Platt calibration (pooled held-out area under the receiver operating characteristic curve, ROC-AUC, 0.659). In every fly-off the planner uses the model trained on the other three terrains, never on the terrain it flies over.

Altitude learner. Because the model cannot choose altitude (Section V-C), altitude is learned during the mission. The learner holds an incumbent altitude and rewards each altitude by the fraction of robots with C2 up. Its estimates are time-blocked (240 s blocks), so only comparisons made close in time count. Every few decisions it probes a neighbouring altitude (±5 or 10 m), and it moves when the lower confidence bound of the neighbour-minus-incumbent difference is above zero ($z = 0.5$). The *cautious* variant adds a home altitude, the hover altitude of 40 m: a move further from home must clear 1.5 standard errors *and* a 0.02 gain, a move back toward home only the normal bar, and probes are rarer (every 4th decision, every 20th once all neighbours are worse).

D. Delay-Adaptive Uplink Cap

The cap moves the bottleneck from the UE into the robot's kernel, where it can be bounded and prioritised, and lets its rate follow the link. It needs no change to the RAN or core. On each robot's TUN interface, a hierarchical token bucket (HTB) shaper has a ceiling equal to the cap c. C2 and ping (ICMP) traffic use a high-priority class with a 20-packet queue; video uses a low-priority class with a 10-packet queue, so excess video is dropped rather than buffered. Each robot pings the core every 0.2 s. ICMP bypasses the kernel video queue, so its round-trip time (RTT) rises only when the UE's own buffer backs up. With r_0 the minimum RTT over 60 s and $\tilde{r}$ the median over the last second, Algorithm 1 runs every 0.2 s. The "in use" test keeps an idle robot's cap from rising and then overloading the link when video resumes. The controller re-creates the shaper when a UE reattaches with a new TUN. Its only inputs, RTT and bytes sent, are available on real hardware.

Algorithm 1 Per-Robot Uplink Cap (Every 0.2 s)

initially $c \leftarrow 200$ kb/s, *slow* $\leftarrow$ true
if $\tilde{r} > r_0 + 150$ ms **or** no reply for 1 s **then**
  $c \leftarrow \max(50,\ 0.7c)$; *slow* $\leftarrow$ false; hold 2 s
else if $\tilde{r} < r_0 + 50$ ms **and** sent $\geq 0.7c$ **and** 2 s since last raise **then**
  $c \leftarrow \min(1600,\ \textit{slow}\ ?\ 2c : c + 50)$
end if

TABLE II
PLACEMENT ARMS

Arm	x–y	Altitude	Role
A	held	fixed 40 m	baseline (hover)
B	robot centroid	fixed 40 m	baseline (proposal)
C	robot centroid	learned on the spot	learned altitude
D	risk model	learned on the spot	learned x–y–z

IV. EXPERIMENTAL SETUP

Traffic and metrics. Each robot sends constant-rate UDP (User Datagram Protocol) video at 1.5 Mb/s to the core and is probed for C2 by ping; a robot's C2 is up when fewer than 50 % of its probes are lost. These definitions differ from those of [5] (two robots, 8 Mb/s video bursts, C2 up when any of five echoes returns), so numbers are not comparable across the two papers. Placement runs are reported as the time-weighted share of a mission with mean C2 up, with all four robots up, and with no robot up, and the 90th-percentile video latency. Forecasts are scored by the area under the precision–recall curve (PR-AUC) relative to the no-skill rate (lift) and by ROC-AUC, pooled over held-out terrains and on the worst held-out terrain.

Data. The forecasting and altitude-effect results use 54,572 robot-cycles from 20 missions of 90–200 min over four terrains (open field, urban, forest, hills), all flown before the cap existed. The cap was tested in a static load test in forest and open field (6 min of video between 2-min no-load phases), in a 30-min forest patrol, and in the placement fly-offs.

Placement arms. Every arm flies the same pilot, seed and 90-min mission and differs only in how x–y and altitude are chosen (Table II): the pilot decides every 30 s, changes altitude by at most 10 m per decision, and stays within a 20–60 m geofence. The 40-m hover altitude and the 20–60 m band are those of [5]: 40 m is the middle of the band and clears every obstacle in the outdoor worlds. Arm B, the robot centroid at fixed altitude, is the baseline of our research proposal; arm A, fixed hover, is the simplest policy; the repositioning policy of [5] is a further baseline for the risk planner.

Validation protocol. A result is accepted only after it survives the control that matches its threat: leave-one-terrain-out evaluation with worst-fold reporting for forecasting; censoring of dead-process cycles for every label; a same-policy null control (the same comparison applied to two nights of one policy) for closed-loop comparisons made across nights, where a comparison counts only if the null control's interval rules out a swing as large as the effect; and the uplink cap for any forest

TABLE III
STATIC FOREST LOAD TEST, FOUR ROBOTS, 6 MIN OF VIDEO

	No cap	Fixed 100 kb/s	Adaptive
Uplink ping loss, video on	64–91 %	0 %	0 %
Uplink ping RTT, median	3–4 min	69–87 ms	76–108 ms
All probes lost (on / off)	64 / 100 %	0 / 0 %	0 / 0 %
Recovery after video	14–18+ min	none	none
Video goodput (Mb/s)	0.13–0.49	0.10	0.10–0.42
Video latency, median	2.2–2.8 min	1.08–1.11 s	0.30–1.08 s

outcome. Confidence intervals are 95 % block bootstraps over contiguous stretches (10-min blocks for missions), because consecutive cycles are strongly correlated; an effect whose interval includes zero is reported as not established. Placement arms are flown on several seeds per world and compared by a two-stage bootstrap that resamples runs and then blocks within each run, so the interval includes run-to-run variation. Before these runs we fixed the rule that a learned arm beats a baseline only if that interval excludes 0. We call an arm equivalent to hover when the 90 % interval of the difference lies within ± 5 points; this margin was set after the first two seeds were seen.

V. RESULTS

A. C1: Uplink Queuing, Not Coverage, and an Autonomic Cap

The long forest outages are a queue inside the UE. With the four robots parked under canopy (24–30 dB uplink attenuation) and the UAV at 40 m, C2 was clean without load. Once the four 1.5 Mb/s video feeds started, uplink ping RTT grew from about 60 ms to minutes within the first seconds and kept rising for the whole load period, peaking at 350–444 s (Fig. 2(a)); 64–91 % of pings were lost and, for 64 % of the time, all eight probes (one per direction per robot) were lost at once (Table III). Probes were delayed rather than dropped, and the links drained only 0.13–0.49 Mb/s of the 1.5 Mb/s offered. The kernel queue on each robot's TUN interface held no backlog and dropped nothing, so the excess was buffered beyond the kernel, inside the OAI UE software (E1); we did not instrument which UE layer holds it. The drain rate is not simply what the radio supports: in the capped placement runs the gNB's own counters (modulation, code rate, allocated symbols) implied full-carrier uplink rates of tens of Mb/s even in forest, far above 1.5 Mb/s, and the emulator's radio thread ran near one full core. Whether the limit is scheduling or emulation is under measurement. [TODO: capacity sweep 0–35 dB and 1/2/4-robot saturation, with gNB MAC statistics and per-thread CPU] After the video stopped, C2 stayed dead for 14 to more than 18 min, consistent with each link slowly draining its backlog.

Bounding the queue removes the outages. A fixed 100 kb/s cap below every link's drain rate, applied at the same placement, removed all C2 loss and the recovery tail, but discarded most of the video (Table III). The adaptive cap of Section III-D also gave zero C2 loss while settling at a different rate on each link (medians 105–429 kb/s, 32–40 cuts per robot in 6 min), delivering 0.10–0.42 Mb/s of video at a median latency of

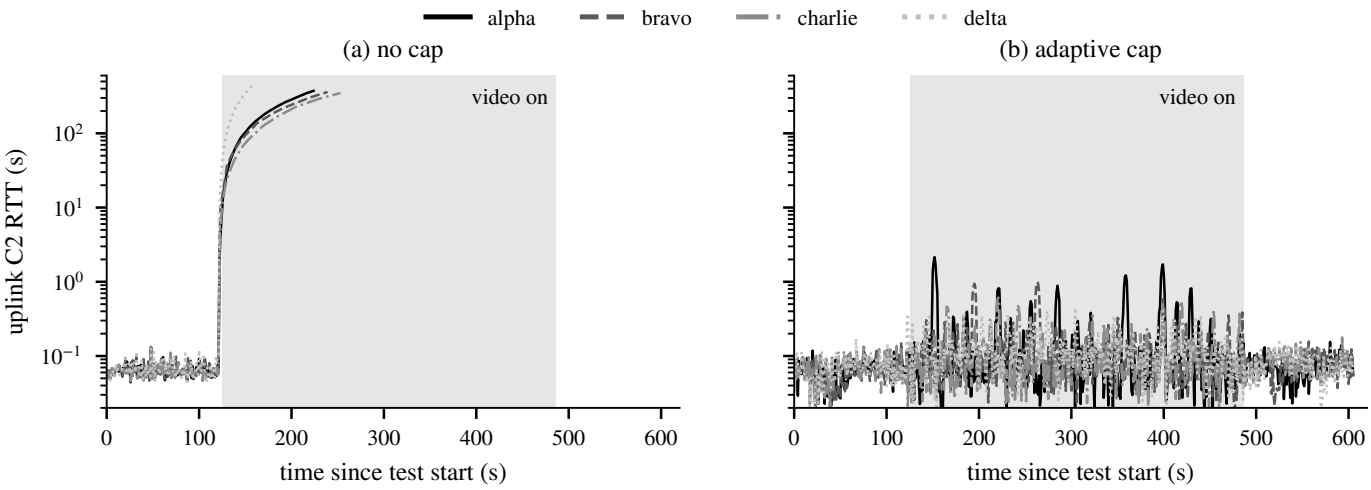


Fig. 2. Uplink C2 ping RTT per robot in the static forest load test (log scale; video on in the shaded interval). (a) Without a cap, RTT grows to minutes; traces end where replies stop before the test ends. (b) With the adaptive cap, RTT stays near its unloaded level.

0.30–0.42 s on three robots; the fourth, with the weakest link, ran at 1.08 s, the time its 10-packet video queue takes to drain at about 100 kb/s (Fig. 2(b)). In open field, with 2–9 dB of attenuation, every cap rose to its 1.6 Mb/s ceiling, made no cuts, and carried the full 1.55 Mb/s per robot at 82–89 ms median latency (E4): in this test, where links were strong, the cap stayed out of the way.

The cap lifts the like-for-like placement arms. In a 30-min forest patrol with the UAV hovering at 40 m, the cap raised mean C2 availability from 66.8 % to 91.8 % against the same seed uncapped, cut the time with no robot connected from 17.4 % to 0.9 %, and the 90th-percentile video latency from 73 s to 1.6 s (Table IX). In the forest fly-off (Fig. 4), only arms A and B flew the same policy with and without the cap: there, the time with no robot connected fell from 25.4 and 25.9 % to 0.3 and 0.0 %, and mean C2 rose from 66.6 and 58.0 % to 89.1 and 88.9 %. Arms C and D also changed altitude learner between the two conditions, so their improvement cannot be attributed to the cap alone. The repeat campaign (Section V-D) adds capped runs of A and B. All four robots were up 84.8 and 85.6 % of the time in open field (two runs each) against 50.2 and 57.0 % uncapped, and 70.6 % (three runs) and 61.1 % (two runs) in forest against 58.4 and 42.8 %. The uncapped side is still one run per arm on another night. In open field every capped run (82.5–87.1 %, all arms) lies above the upper end of both uncapped runs' intervals (61 and 65 %), and the same-policy null control there is -0.5 [-4.3, $+3.0$] (Table VII), so a gain of about 30 points is not a night-to-night swing. In forest the null control is too wide to rule out the gain's size.

The cap turns minute-long outages into single missed cycles. For the same like-for-like runs, Fig. 3 gives the length of every C2 outage. Without the cap the median forest outage lasted 143 s (arm A) and 97 s (arm B), and 64 % and 59 % of outages exceeded a minute; with it the medians fell to 17 s and 19 s and 11 % and 19 % exceeded a minute. In open field the median fell from 48 s to 11 s (Table X). Capped forest

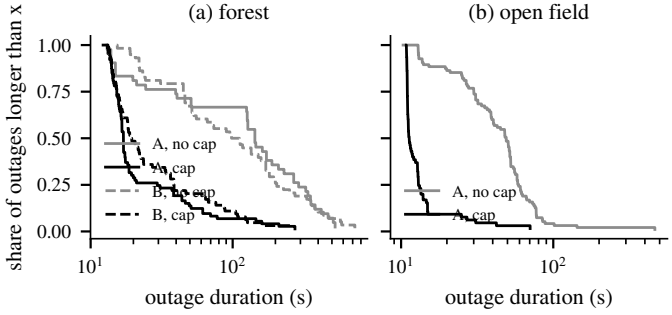


Fig. 3. Length of C2 outages (time a robot's C2 stayed down), like-for-like arms with and without the cap, one 90-min run each: share of outages longer than a given duration. Durations below about 15 s are not resolved.

runs had somewhat more outages per robot-hour (12.2 against 7.0 for arm A), almost all a single measurement cycle long; at 11–20 s per cycle, that is the shortest outage the logs can resolve.

The same mechanism may bear on the forest result of our earlier study [5], where a roaming node drove 8 Mb/s video bursts to over 70 % loss; its traffic pattern differed, and we have not tested this.

Cap versus 5G-native separation. A separate network slice or QoS flow for C2 would also keep C2 out of the video queue, but not shorten the video's own delay. [TODO: slice-per-traffic-class experiment: no cap, cap, slice, slice + cap, static forest load test and one patrol]

B. C2: Terrain-Blind Forecasting Has Modest Skill, and Capacity Hurts

A linear model forecasts failure onset 20 s ahead on unseen terrain, weakly. After dead-process censoring (Section III-B), the best model, an L1-regularised logistic regression on 65 terrain-blind features, reaches a pooled PR-AUC 1.79 times the no-skill rate and ROC-AUC 0.667, and no held-out terrain

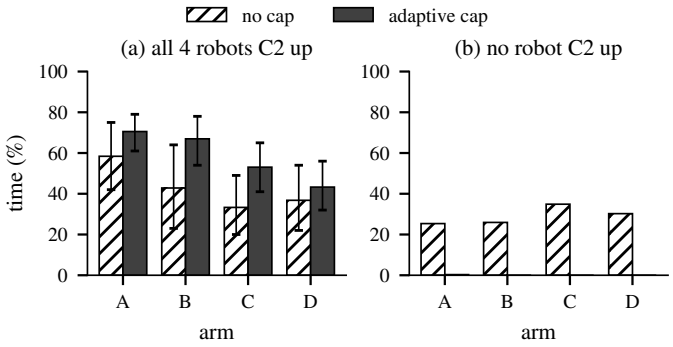


Fig. 4. Forest fly-off, each placement arm without and with the adaptive cap (one 90-min run per bar; whiskers: 95 % block-bootstrap CI). Arms C and D used an upper-confidence-bound (UCB) altitude bandit uncapped and the prober learner capped, so only A and B flew the same policy in both conditions.

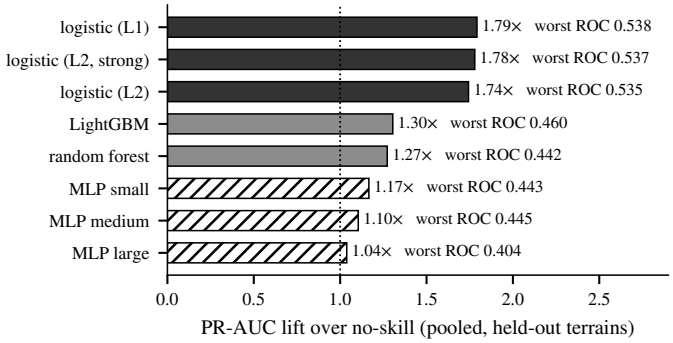


Fig. 5. Forecasting skill by model class: pooled PR-AUC lift over no skill on held-out terrains, and the worst held-out-terrain ROC-AUC (65 features, 20-s onset horizon, 38,303 onset observations, leave-one-terrain-out).

falls below ROC-AUC 0.538 (Fig. 5). Skill peaks at a 20-s horizon: a six-feature trend model scores ROC-AUC 0.650 at 20 s and 0.552 at 60 s (Table VI). Forest labels in this data predate the cap and are partly queue backlog (Section V-A). [TODO: retrain all eight model classes on the capped, monitor-protected runs and report beside these numbers]

More capacity is worse. On identical folds and features the ordering is monotone in capacity: three linear models lead, tree ensembles follow, and three multilayer perceptrons trail. The largest network scores ROC-AUC 0.503, and 0.404 on its worst terrain, below chance. One exception should be stated: on six trend features at a 60-s horizon, the medium network beats the linear models on pooled lift (1.55 against 1.21–1.22) but not on its worst terrain (ROC 0.456 against 0.550). Sequence models on 10 Hz radio traces add nothing: on 3,115 labelled cycles over three folds, a temporal convolutional network (TCN) and a gated recurrent unit (GRU) average ROC-AUC 0.493 and 0.523 against 0.540 for a logistic regression on window summaries (A7). Under leave-one-terrain-out validation with four terrains, we take worst-fold skill, not pooled skill, as the figure of merit; our reading is that it rewards models that learn a direction rather than thresholds tied to the training terrains.

TABLE IV
ALTITUDE-EFFECT CONTRASTS (C2 AVAILABILITY PER +10 M): MEASURED, AND RECOVERED FROM A TERRAIN-BLIND LIGHTGBM MODEL AT TWO FORECAST HORIZONS. *: 95 % BLOCK-BOOTSTRAP CI EXCLUDES 0

	measured		recovered	
Contrast	Cycle	60-s bins	h=20 s	h=60 s
Open − forest	$+0.208^*$	$+0.167^*$	$+0.001$	$+0.006$
Open − hills	$+0.153^*$	$+0.138^*$	-0.003	$+0.037^*$
Urban − forest	$+0.073^*$	$+0.070^*$	$+0.006$	-0.020^*
Hills − forest	$+0.055^*$	$+0.030$	$+0.003$	-0.031^*

C. C3: The Model Does Not Recover How Terrain Changes the Altitude Effect

The measured effect of climbing differs strongly between terrains: per +10 m, C2 availability gains 0.208 more in open field than in forest per cycle, and 0.167 more on a uniform 60-s time base that removes the outcome-dependent cycle length (Table IV). This terrain dependence was reported in [5]; the question here is whether a terrain-blind model recovers it. To give the model every chance we use a gradient-boosted tree model (LightGBM), which can represent interactions between altitude and link state, trained on the same terrain-blind features and held-out terrains. It does not recover it. At the 20-s horizon of Section V-B, the open-field-minus-forest contrast in the model's risk is $+0.001$ ($p = 0.89$), and no contrast is significant. At a 60-s horizon it is $+0.006$ ($p = 0.37$), and two other contrasts are significantly inverted. On coarser 120-s bins one recovered contrast reaches significance but stays about one sixth of the measured value (Appendix A).

The consequence for control is direct. Replayed through the planner on each held-out terrain, no surrogate chooses altitude sensibly: linear models choose about 55 m in every terrain, and a gradient-boosted model chooses about 47 m in both forest and open field (D1). Altitude therefore has to be learned during the mission.

D. C4: Learned Placement Under the Cap

An exploring altitude learner did worse than the baselines in single runs. In the first capped forest fly-off (seed 1), hover (A) and centroid (B) kept all four robots connected 70.5 % and 67.0 % of the time, indistinguishable from each other (B−A -4.0 [-20.4, $+11.2$] points). The two arms with an exploring learner were worse within their runs: C−A -18.1 [-34.1, -2.6] and D−A -27.4 [-43.0, -10.7]. These are single runs on different nights, and the night-to-night swings of a same-policy null control reach about 20 points (Table VII), so the size of the loss is not established. Both exploring arms spent much of the mission at 45–55 m (Fig. 6).

One mission cannot determine the sign of the forest altitude effect. Comparing arm C with arm B at matched times, with the same x–y rule and seed, gives -6.6 points of C2 per +10 m ($[-14.9, +0.9]$): probably harmful, not established, and too small on its point estimate to explain the whole gap above. The slope the learner itself can estimate within one 90-min run

is $+4.6$ ($[-9.1, +12.0]$) in arm C and $+5.9$ ($[-10.7, +14.9]$) in arm D. Its interval spans both signs, so the learner moved on noise.

Repeated missions: no learned arm beats hover. We flew every arm with the cap, and the cautious learner for C and D, on three seeds in each world, 24 missions in all (Fig. 7). Two start-ups that failed before any data, and one run stopped by a host memory fault, were flown again and the failed attempts excluded. Because the runs fall on different nights and a seed does not reproduce the robots' paths, arms are compared by a two-stage bootstrap (resample runs, then 10-min blocks within each run). The rule was fixed before these runs: a learned arm beats a baseline only if its interval excludes 0. No arm does. In open field every arm is equivalent to hover, which keeps all four robots up 84.2 % of the time (90 % intervals within ± 5; 95 % intervals given here): B$-$A $+1.2$ [-0.9, $+3.5$], C$-$A $+1.0$ [-3.4, $+4.5$], D$-$A -0.7 [-5.6, $+3.3$]. Placement has nothing left to win there once the queue is bounded. In forest, hover is the steadiest arm (67.9–73.2 %, mean 70.6 %) and the arms that move the UAV trail it: B$-$A -8.4 [-19.4, $+2.1$], C$-$A -7.0 [-19.9, $+4.7$], D$-$A -8.0 [-18.6, $+4.2$]. None of these gaps is established. Each moving arm had one run 16–18 points below hover's run on the same seed, and others close to it.

Climbing and moving cost the links in forest. The cautious learner's lowest run (52.1 %) is the one that left its 40-m home for long: it spent 59 % of the mission at 45 m or higher, against 7–11 % in its other two runs, and in that run the number of robots connected fell as altitude rose (correlation -0.17). This fits the negative capped slope above but does not establish it. Moving the UAV also made the cap work harder. Averaged over forest runs, the cap cut a robot's rate 266 times per hour under hover and 364 under the planner, video goodput fell from 1.03 to 0.87 Mb/s, and 90th-percentile video latency rose from 1.5 to 2.2 s. Each move changes the robots' paths through the canopy, and the cap has to find their new rates. Horizontal UAV–robot distance does not explain a robot's C2 in forest (correlation 0.00 over 48 robot-runs): what matters is where a robot stands relative to the trees, which the geometry features cannot see (C3). In effect the cautious learner is hover with short probes: it gave no benefit over hover, and when it did leave 40 m for long, it lost.

The risk planner survives its null control in open field only. Against the repositioning policy of [5], matched on geometry, the planner gained $+10.5$ [$+7.0$, $+13.2$] points in open field. The same comparison between two nights of the *same* baseline policy gave -0.5 [-4.3, $+3.0$] there, which rules out a night-to-night swing of that size, so the gain is interpretable. In forest the planner lost -15.3 [-24.8, -5.0], but the null control's interval [-17.6, $+6.4$] admits a swing that large; in urban and hills the null was itself as large as the effect ($+18.7$ and -21.7). Those three comparisons are uninterpretable (Table VII).

VI. Limitations and Threats to Validity

The radio is real 5G software, but propagation is injected: the channel has large-scale loss and correlated shadowing but no multipath, terminal height is fixed at 0.5 m, and absolute availability is internally consistent rather than externally calibrated. The OAI UE may buffer uplink data differently from a commercial modem, and constant-rate video is a worst case for the queue; whether C1 transfers is the first question for hardware. Every between-terrain claim rests on four terrains. The forecasting and altitude-effect data predate the cap, so their forest labels are partly queue backlog, and the dead-process rule removes only about half of the dead cycles the monitor recorded. Throughput was not measured with active probes during the campaigns, because they would have congested the links under test; the emulator's radio thread, near one full core, may cap the cell's total uplink [TODO: measured capacity from the capacity sweep]. Placement comparisons rest on three 90-min runs per arm and world (one for the exploring-learner and uncapped arms), so forest gaps of up to about 10 points are not established, and the capped and uncapped runs of the same arm were flown on different nights. The robots' safety guards react to link state, so the same seed does not give identical patrols across arms. Appendix C gives the full list.

VII. Conclusion

We built a three-loop management stack for a UAV-borne 5G cell serving four search-and-rescue robots and flew it in closed loop over real 5G software. The most effective loop was not learned: a per-robot uplink cap driven by round-trip time removed a silent queueing failure inside the UE, shortened C2 outages from minutes to single measurement cycles, and improved both placement arms flown with and without it (C1). The learning layer forecasts link-failure onset 20 s ahead with modest skill that falls as model capacity grows (C2), does not recover how terrain changes the effect of altitude (C3), and, over repeated closed-loop missions, does not beat fixed hover; in forest every arm that moved the UAV trailed it, and a cautious learner helped only by staying near hover (C4). For autonomous management of such cells, the lesson is to verify that the managed system is measured and loaded soundly, bounding its queues and checking that its processes are alive, before learning where to fly it. The next steps are a hardware trial of the cap and retraining the layer on capped, watchdog-protected data.

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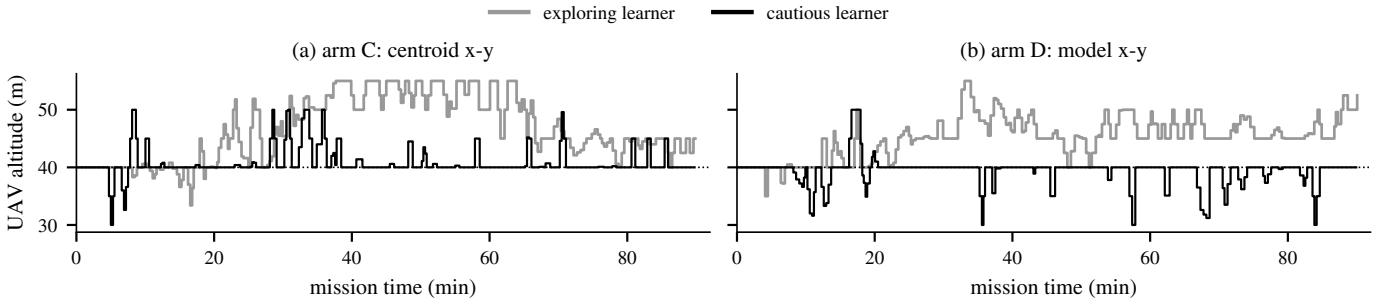


Fig. 6. UAV altitude during the capped forest missions for the two learned-altitude arms. The exploring learner drifts to 45–55 m; the cautious learner keeps 40 m as its incumbent and leaves it only for short probes.

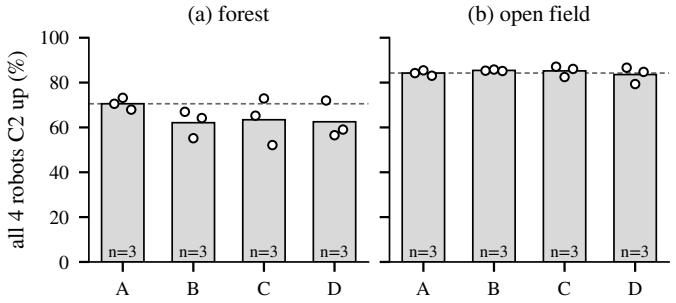


Fig. 7. All four robots C2 up per arm with the cap, cautious learner for C and D: bar = mean over runs, dots = individual 90-min runs, dashed line = hover's mean. Three runs per arm and world. Open field is tight; forest varies from run to run.

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APPENDIX A
EXTRA TABLES

TABLE V
DEAD-PROCESS CONTAMINATION BY TERRAIN (6,760 OF 54,572 ROWS, 12.4 %)

Terrain	Dead share of C2-down	C2 raw	C2 censored
Hills	72.2 %	83.0 %	94.6 %
Forest	53.4 %	37.9 %	56.7 %
Urban	39.0 %	69.6 %	79.0 %
Open field	3.9 %	78.5 %	79.2 %

TABLE VI
FORECAST HORIZON SWEEP, CENSORED ONSET POPULATION, POOLED OVER HELD-OUT TERRAINS

Horizon	GBM lift	GBM ROC	Logit-6 lift	Logit-6 ROC
10 s	1.44	0.569	1.42	0.575
20 s	1.29	0.558	1.52	0.650
30 s	1.19	0.544	1.38	0.625
60 s	1.23	0.579	1.22	0.552

TABLE VII
RISK PLANNER VERSUS THE REPOSITIONING BASELINE OF [5], WITH A SAME-POLICY NULL CONTROL (GEOMETRY-MATCHED, 95 % CI)

Terrain	Planner − two_phase (pp)	Null control (pp)	Usable
open field	+10.5 [+7.0, +13.2]*	−0.5 [-4.3, +3.0]	yes
forest	−15.3 [-24.8, -5.0]*	−5.0 [-17.6, +6.4]	no
urban	+22.6 [+18.4, +27.2]*	+18.7 [+12.9, +24.4]*	no
hills	+26.3 [+19.4, +32.5]*	−21.7 [-32.1, -12.2]*	no

TABLE VIII
OFFLINE ALTITUDE BENCH, FRESH SEEDS: LEARNER MINUS FIXED 40 M (REWARD POINTS)

Traces	Learner	Worst mean	Runs < -2	Climb gain
All, noise 0	exploring	−1.0	3.7 %	+18.9
	cautious	−0.6	1.2 %	+9.8
40-m, noise .03	exploring	−1.2	11.3 %	+6.5
	cautious	−0.7	3.1 %	+2.5
Capped forest, .03	exploring	−1.9	22.8 %	+2.1
	cautious	−0.8	4.4 %	+0.6

Dead-process rule against the liveness monitor. Over C2-down cycles in the 29 robot series where the monitor wrote events, the rule flagged 2,583 cycles, the monitor recorded 3,878 dead or relaunching ones, and 2,189 were both: precision 0.85, recall 0.56. The other 4,177 flags lie in series where the monitor had stopped and cannot be checked (`dead_process_validation.csv`).

Recovered contrasts on uniform time bases. For the 60-s-horizon model, the open-field-minus-forest contrast on 30-, 60- and 120-s bins is -0.001, $+0.010$ and $+0.021$ (the last significant, $p = 0.017$), against measured $+0.184$, $+0.167$ and $+0.117$.

TABLE IX
FOREST PATROL, FIRST 30 MIN, HOVER AT 40 M, SAME SEED

	No cap	Adaptive cap
Mean C2 up	66.8 %	91.8 %
All 4 up (95 % CI)	52.0 % (16–76)	72.7 % (67–77)
No robot up	17.4 %	0.9 %
Video latency median / p90	0.06 / 73.4 s	0.11 / 1.6 s

TABLE X
C2 OUTAGE LENGTHS, LIKE-FOR-LIKE ARMS WITHOUT AND WITH THE CAP (ONE 90-MIN RUN EACH)

Run	Cap	Outages/robot-h	Median (s)	p90 (s)	Max (s)
forest A	no cap	7.0	143	378	526
forest A	cap	12.2	17	59	275
forest B	no cap	9.7	97	385	725
forest B	cap	10.7	19	88	259
open field A	no cap	15.9	48	77	465
open field A	cap	10.9	11	15	70

APPENDIX B
REPRODUCIBILITY

[TODO: Step 9: map every result to script, data file and command]

APPENDIX C
THREATS TO VALIDITY IN FULL

Channel. Every run uses an additive white Gaussian noise channel in the OAI radio simulator; the bridge injects large-scale loss and distance-correlated shadowing only, so there is no small-scale fading, Doppler or frequency selectivity. Terminal height is a constant 0.5 m, there is no two-ray, Fresnel or rubble term, the line-of-sight sigmoid uses the urban coefficients in every world, and the applied attenuation is clamped at 45 dB. The smooth channel may itself limit what a short-horizon forecaster can learn.

Sampling. Each measurement cycle waits for its probes, which block when the link fails, so cycle length depends on the outcome (longer when C2 is down, and longest in forest). Altitude effects are therefore also reported on uniform time bases (Table IV); the model-recovered contrast at 120 s is significant but small, and we report it.

Labels. Dead-process cycles are censored by a rule, not by ground truth; the forecasting and altitude-effect data predate the uplink cap, so forest failure labels are partly queue backlog, which the 20-s forecaster cannot see.

Sample size. Four terrains bound every between-terrain claim; the indoor world produced no usable runs. The 10 Hz radio traces give 3,115 independent labelled cycles over three folds.

Closed-loop comparisons. The exploring-learner and uncapped runs are one 90-min run per arm on different nights, with within-run block-bootstrap intervals. The repeat campaign has three runs per arm and world, and its intervals resample runs as well as blocks, which with so few runs gives only a rough interval. The robots' safety guards react to link

state and consume the random stream, so a shared seed gives similar but not identical patrols. Comparisons across nights are accepted only where a same-policy null control's interval rules out a swing as large as the effect (Table VII); the fly-off arms have no null control, so their cross-night differences are not treated as established.

Generality of C1. The UE buffering behaviour is that of the OAI UE; a commercial modem may bound or drop uplink data. Constant-rate video is a worst case; an adaptive encoder would congest less but reacts more slowly than the cap. The relevance to the forest result of [5] is untested.